

Computer Science & Information Technology

Computer Organization & Architecture

DPP: 1

Floating Point Representation

Q1 Which of the following is the representation of $(-1)_{10}$ in IEEE-754 single precision floating point number?

- (A) $S = 1, E = 00000000, M = 000000000000000000000000$
 (B) $S = 1, E = 01111111, M = 000000000000000000000000$
 (C) $S = 1, E = 10000000, M = 000000000000000000000000$
 (D) $S = 1, E = 01111111, M = 100000000000000000000000$

Q2 Which of the following is the representation of $(0.0000101)_2$ in IEEE-754 single precision floating point number?

- (A) $S = 0, E = 01111010, M = 101000000000000000000000$
 (B) $S = 1, E = 01111010, M = 101000000000000000000000$
 (C) $S = 0, E = 01111010, M = 010000000000000000000000$
 (D) $S = 0, E = 01111011, M = 101000000000000000000000$

Q3 The value of a float type variable is represented using the single-precision 32-bit floating point format IEEE-754 standard that uses 1 bit for sign, 8 bits for biased exponent and 23 bits for mantissa. A float type variable X is assigned the

decimal value of -22.25 . The representation of X in hexadecimal notation is

- (A) C1B40000H (B) 41B20000H
 (C) C1B20000H (D) 41B40000H

Q4 Consider the following representation of a number in IEEE 754 single-precision floating point format?

0 10000011 110000000000000000000000

The decimal value corresponding to the above representation is ____ ?

Q5 Minimum possible positive normalized value represented in IEEE-754 single precision format is?

- (A) $S = 0, E = 00000000, M = 000000000000000000000000$
 (B) $S = 0, E = 00000001, M = 000000000000000000000000$
 (C) $S = 0, E = 00000000, M = 100000000000000000000000$
 (D) $S = 1, E = 00000001, M = 100000000000000000000000$

Q6 Maximum possible positive denormalized value represented in IEEE-754 single precision format is?

- (A) $(2^{23}-1) \cdot 2^{-150}$
 (B) $(2^{24}-1) \cdot 2^{-149}$
 (C) $(2^{23}-1) \cdot 2^{-149}$
 (D) $(2^{24}-1) \cdot 2^{-150}$



Answer Key

Q1 (B)

Q2 (C)

Q3 (C)

Q4 28~28

Q5 (B)

Q6 (C)



[Android App](#) | [iOS App](#) | [PW Website](#)

Hints & Solutions

Q1 Text Solution:

for $(-1)_{10}$, $\text{sing} = 1$ for negative value

$$(1)_{10} = 1.0 \times 2^0$$

Mantissa = 000 0

Original exponent = 0

Bias exponent = $0 + 127 = (127)_{10} = (01111111)_2$

Q2 Text Solution:

For positive value sign $S = 0$

$$(0.0000101)_2 = 1.01 \times 2^{-5}$$

Mantissa = 01000 0

Original exponent = -5

$$\text{Biased exponent} = -5 + 127 = (122)_{10} = 01111010$$

Q3 Text Solution:

For negative value sign $S = 1$

$$(22.25)_{10} = (10110.01)_2$$

After implicit normalization = 1.011001×2^4

Mantissa = 011 00 1000 ... 0

Original exponent = 4

Biased exponent = $4 + 127 = (131)_2 = (10000011)_2$

The number will look like as:

1	10000011	01100100000000000000 00000
---	----------	-------------------------------

S	E	M
---	---	---

In hexadecimal the number will be: C1B20000

Q4 Text Solution:

Sign is 0, hence it's a positive value.

Given biased exponent = $(10000011)_2 = 131$

$$\text{Value} = +1.11 \times 2^{131-127}$$

$$= 1.11 \times 2^4$$

$$= (11100)_2$$

$$= +28$$

Q5 Text Solution:

If normalized value then it will be having biased exponent $(1)_{10}$. And mantissa can have all zeros because number will be implicitly normalized.

Q6 Text Solution:

For positive number sign $S = 0$

For denormalized number $E = 00000000$

Mantissa should be all (23 times) 1s.

$$\text{Value} = + 0.1111...1 * 2^{-126}$$

$$= 111111111111111111111111.0 * 2^{-23} * 2^{-126}$$

$$= 111111111111111111111111110 * 2^{-149}$$

$$= (2^{23}-1) \cdot 2^{-149}$$



[Android App](#) | [iOS App](#) | [PW Website](#)